Proceedings of the Third International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT-2025)
IEEE Xplore Part Number: CFP25CV1-ART; ISBN: 979-8-3315-2754-9

2025 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT) | 979-8-3315-2754-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/IDCIOT64235.2025.10915120

Smart Sales Forecasting Machine Learning Models for Demand Prediction in Retail

Sandeep V
Computer Science and Engineering
Chennai Institute of Technology
Chennai, India
sandeepv.aiml2023@citchennai.net

Dr. Kandavel N
Computer Science and Engineering
Chennai Institute of Technology
Chennai, India
kandaveln@citchennai.net

R Saivenkat
Computer Science and Engineering
Chennai Institute of Technology
Chennai, India
rsaivankat.aiml2023@citchennai.net

Dr.Gowri R
Computer Science and Engineering
Chennai Institute of Technology
Chennai, India
gowri.r@citchennai.net

Thirumalai Murugan R
Computer Science and Engineering
Chennai Institute of Technology
Chennai, India
thirumalaimuruganr@citchennai.net

Zahid Hussain J
Computer Science and Engineering
Chennai Institute of Technology
Chennai, India
zahidhussainj@citchennai.net

***Abstract*— Forecasting accurately the upcoming product demand is important to the retail sector and is achieved by the timely use of stock and sales strategies. The study investigates how a machine learning model based on all the possible external variables other than the traditional historical sales data such as weather conditions, fuel prices, and previous year sales records can be used to predict sales trends. By the utilization of these external and time variables, the model comes up with a complete prognosis method, which is to be matched with the uncovering of possible patterns that conservative techniques might not identify. Examine a variety of machine learning methods, including time series analysis and regression models, to ascertain the technique that is the most suited to the study of the complex influences of consumer behaviour. Our findings show that by accounting for the external data from the economic field and environment that is related to the area, the accuracy of the forecast along with the adaptability of sales targeted at the involved company increase significantly, thus making it a very powerful tool. This predictive model not only enhances decision-making for inventory control but also enables retailers to better anticipate demand swings that are caused by such external factors, thus new merit is shared in strategic planning and operational efficiency.**

Keywords— ***Sales Prediction, Retail Analytics Machine Learning, Historical Sales Data, Predictive Modelling.***

I. Introduction

In the fast-changing retail world, good sales forecasting is important for inventory management, coordination with market demand, and strategic business decision-making. The conventional approach to predicting sales is to lean mainly, on the one hand, on the past sales records which often miss incorporating the factors, for instance, the changing external environment, in informing consumer purchasing behaviour. This research applies the machine learning technique in order to make a more accurate sales forecast based not on a number of variables such as weather conditions, fuel prices, and economic indicators together with the previous year's sales results. From large data, helps in processing the data, analyzing patterns and learning from various sources. Our model which includes external variables has been operably holistic, therefore taking into account necessity foresight considerations variables such as seasonality, economy, and environmental impact along with customer needs that could be negatively or positively stimulated by these three factors. The proposed model within this study integrates multiple algorithms of machine learning and analyses their predictive performance in terms of capturing the sales trends affected by diverse factors. Therefore, expected that by using this methodology, including data on economic and environmental activities along with historical records may yield more accurate and adaptive predictions of sales. Improvements in the field of forecasting can help retailers prevent stockouts, lower inventory costs, and manage to have the right quantities in the stores at appropriate time for consumers and thereby improve their revenues in business.

II. Predictive Analysis Using Hierarchical Agglomative Clustering

A. *Advancement made in Sales Prediction*

With the advancement of the concepts of artificial intelligence and machine learning, there have a been quite a bit of studies and research conducted by many people in various fields including predictive analysis in a lot of domains in the industries like retail, healthcare, finance, and marketing. The studies done so far have been focused on predictive analysis using different algorithms, methodologies and approaches.

A study was conducted by Ruitenbeek, Koole, Bhulai [1] were done product sales forecasting using a hierarchical clustering algorithm. The emphasis on clustering algorithms for sales forecasting has been laid fluently by the authors using data and the insights gained from the data. How is clustering better: Improved Forecast Accuracy with Clustering, Effectiveness for Intermittent Demand, Comparative Analysis with Traditional Grouping, Scalability and Robustness. The study utilizes time series data from over 3,000 unique products, focusing on daily sales data for a Dutch retail company specializing in bicycle accessories. This time series data includes varying demand patterns, such as intermittent demand and high variability in sales volume.

B. *Data Preparation and Preprocessing*

Data Cleaning: The authors remove records with missing critical information (e.g., sales date, product ID) but retain outliers in sales quantities, as these can significantly impact inventory decisions [1].

Handling Product Variability: The dataset includes products with varying sales histories, some spanning only a

Authorized licensed use limited to: Mukesh Patel School of Technology & Engineering. Downloaded on July 10,2026 at 15:48:02 UTC from IEEE Xplore. Restrictions apply.

few days and others many years. To standardize, the authors apply filtering to include only products with a minimum of two years of sales data [2].

C. Forecasting Models

Five models are tested on the individual forecasting approaches to establish a baseline, including naive forecasting models and various configurations of GLMs. GLMs are chosen foy handle binary variables for addressing seasonality and holiday perturbations and permit cases to be non-normally distributed in sales data [3].

Feature Engineering and Evaluation:

Seasonality and Lag Features: Specified characteristics, such as monthly and weekly seasonality, holiday correction, and lag factors, including sales in prior weeks, are designed to identify periodic and recent sales behavior...

Evaluation Metrics: The model's forecasting performance is evaluated using MASE, a scale-independent metric suitable for comparing accuracy across different time series.

D. A Comparative Study of Big mart Sales in Sales Prediction

A study was conducted by Gopal Behera and Neeta Nain on this matter. The dataset that used consisted of the 2013 big mart sales which contained 12 attributes related to Product and outlet characteristics [4]. The data was obtained from a source and the cleaning and exploration processed were carried out by handling the missing values, correcting the unusual and inconsistent values and categorical attributes were adjusted for consistency. Item Visibility values are replaced where appropriate. Categorical Values are standardized, and new attributes are created. Outlet Open Time is included by transforming to an age variable, the number of years since the outlet was established. The models were then developed by applying machine learning techniques with an aim of estimating performances on sales As a result, the various models were tested to ascertain the most suitable ones. The models that used were decision tree, linear regression, ridge regression, and XGBoost.

III. K-Mean Cluster

Cluster-Based Forecasting via K-Means Clustering is the tool makes easier to forecast the demand by grouping similar type of products into clusters. These clusters can then be used for the improvement of the quality of prediction [5,6]. This action begins with K-Means clustering where the items are grouped based on company sales data. The first and important step in forming clusters is choosing K, the number of clusters which may be determined using the Elbow Method. On the identification of the best number of clusters, K-Means assembles the like-demand products and therefore, provides accurate forecasts. Gower's [7] is dissimilar to other clustering algorithms such as Hierarchical Agglomerative Clustering (HAC) [8-10]. Unlike some cluster algorithms, K-Means does not involve the integration functionalities into the classes. However, operates sequentially through a product of the algorithm in charging items to particular clusters, and the shifting of the centroids found in the cluster before arriving at the best disaggregation[11].

As for the types of the similarities measurement, there exist the Cosine Similarity criterion, for instance. This is because the KD tree is better suited to handle time series data then the Euclidean space as do not have the scale-based differences that might develop. Once clusters are formed, the total sales data carried out from each cluster next is employed as a new feature in the prediction of each particular products' volume. The focusing on the trends as the group is the advantage of this method, and these trends are more than the trends that belong to the individual products. This approach not only aids the process of improving the accuracy of the forecasts by weeding out the noise in the individual sales but also gives a better way of storing inventory by predicting the intensity of the demand according to the formation of the different groups of products. In the process of handling large data sets, K-Means clustering is of significant help, since it allows splitting the forecasting procedure into more reasonable parts and, therefore, improves both the scalability and efficiency .

IV. Decision tree Algorithm

This is one of the most widely used methods, which can be applied to both classification and regression problem. The dataset is split based on the features that give the maximum information gain and a model having a treelike structure which predicts is got. The division of the data is based on the entropy and information gain rates which help in making the decision [12]. As in each node, the algorithm chooses that feature that contributes to the maximum reduction of entropy. The entropy in a dataset D is given by:

$$H(D)=-\sum_{L=1}^{n} P_i \log_2(Pi) \quad \text{------ (1)}$$

where p is the proportion of each class in the dataset. The information gain is computed for each feature and the one with the highest gain is selected for the split [13]:

$$IG(D,A)=H(D)-\sum_{v\epsilon A}\frac{|D_v|}{|D|}H(D_v) \quad \text{------ (2)}$$

V. Linear Regression

The statistical technique that determines the relation between a dependent variable y and one or several independent variables $x_1,x_2,\ldots,x_n$[14]. The relationship is modeled by means of a linear equation that is generally expressed as the following:

$$y = \beta_0 + \beta_1x_1+ \beta_2x_2+\ldots\ldots+ \beta_nx_n + \epsilon \quad \text{------ (3)}$$

where:

y dependent variable (target),
$x_1, x_2\ldots, x_n$ independent variables (predictors),
β_0 the intercept,
$\beta_1, \beta_2\ldots, \beta_n$ coefficients of independent variable, and
ϵ error term.

The goal is to minimize the residual sum of squares (RSS) between the predicted and actual values:

Authorized licensed use limited to: Mukesh Patel School of Technology & Engineering. Downloaded on July 10,2026 at 15:48:02 UTC from IEEE Xplore. Restrictions apply.

$$\text{RSS}=\sum_{i=1}^{n}(y_i-\hat{y}_i)^2 \quad \text{------ (4)}$$

Where $\hat{y}_i$ represents the predicted value. Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) are used to measure the accuracy of model.

The Ridge regression is similar to Linear regression, but with an added penalty function to reduce the enormity of weights. These services allowed the addition of the regularization term that ultimately helps to bring the coefficients of less important features closer to zero. The cost function for ridge regression is given by:

$$\text{Cost Function} = \sum_{i=1}^{n}(y_i-\hat{y}_i)^2 + \alpha\sum_{j=1}^{m}\beta_j^2 \quad \text{------ (5)}$$

where:

$\sum_{i=1}^{n}(y_i-\hat{y}_i)^2$ - RSS (similar to linear regression),
α - regularization parameter.
β_j -coefficients estimations in the model.

VI. XGBOOST AND LIGHTBGM

XGBoost is an advanced form of GBM and works better because in XGboost use differentiable loss function.
Algorithm: begins the model, calculates pseudo-residuals in order to fit a base learner, and adjusts the model to reduce error.
Advantages: Compared to other models such as LightGBM, XGBoost supports some more features such as missing data, parallel tree construction, and continue training, which makes the model excellent in terms of both performance and level of accuracy[15].
This paper also included a quantitative analysis of the model in its ability to provide appropriate levels of accuracy and precision with the help of several parameters.

Cross-validation is 20-fold, that means that data are split into 20 parts, and one part is used for testing while other 19 parts are used for training. Performance Metrics: In order to compare the predictive accuracy of models, MAE and RMSE are used and models are said to be evaluated. XGBoost was found to perform best, achieving lower MAE and RMSE in the other models[16-19]. The study concluded by saying that the XGBoost model outperformed the other models making the best fit for the predictive analysis of the big mart sales. Chen and Guestrin [20] proposed XGBoost, a controlled classification technique that gradually improves weak base learners, thus, a better learner can be made. Moreover, the objective function J is defined in this way:

$$J(\Theta) = L(y,\hat{y}) + \Omega(\Theta) \quad \text{------ (6)}$$

where,

(Θ) – Represents the parameters of the model (e.g : Weights in a neural network)
$L(y,\hat{y})$ – The Loss function that is used to measure the difference between the Actual output value (y)and Predicted output value ($\hat{y}$).
$\Omega(\Theta)$ –stands for loss function L2 that used to avoid overfitting.

:

Fig. 1. Comparison Train, Validation Accuracy of XGBoost and GBM

The objective function at t iteration with n instances can be written as:

$$J^{(t)} = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_{k=1}^{t} \Omega f(k) \quad \text{------ (7)}$$

The haste algorithm LightGBM was created by Keetal, in order to reduce the computation time of XGBoost. The primary difference between XGBoost and LightGBM is how these tools grow the decision trees (Fig 1) In LightGBM, the decision trees are improved vertically (leaf-wise) and in XGBoost, improved horizontally (level-wise) [21]. This is an important point to mention that this is what LightGBM is quite good at when it comes to high-dimensional data handling.

Leaf & Level Wise tree growth : -

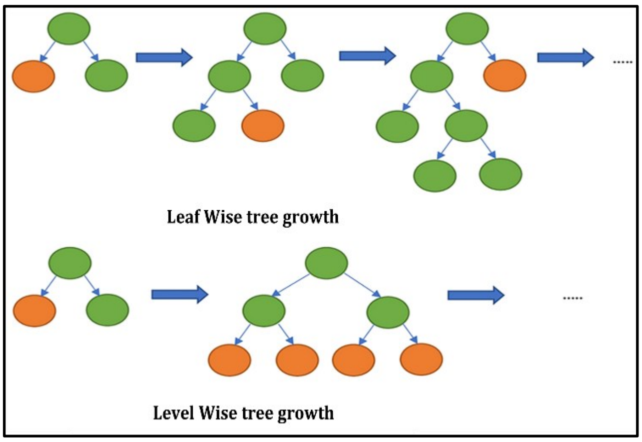

Fig. 2. Comparison between the Leaf Wise tree growth and Level wise tree growth

Authorized licensed use limited to: Mukesh Patel School of Technology & Engineering. Downloaded on July 10,2026 at 15:48:02 UTC from IEEE Xplore. Restrictions apply.

VII. Architecture

According to the proposed system, the prediction of future sales can be achieved by the integration of many data sources, data preprocessing, and the application of machine learning techniques. This is made up of several levels: API and database crawling for data gathering, data preparation for removing noise and formatting the data and, finally, model development to look for cycles and trends. The trained model uses the received elements, including fuel prices and sales in the past, for making the forecasts. The last one is a layer of deployment that provides the simple and functional user interface for making real time predictions and feedback loop for the model improvement. Architecture like this one offers the precision, growth, and quantitative decision-making that is required.

VIII. Scope of Future Study

Future studies on sales prediction would look into integrating other external variables such as geopolitical events and social media trends, in order to enhance the accuracy of the forecasts. Hybrid models combining machine learning techniques, like deep learning, with traditional statistical models may be used to enhance performance in complex sales patterns. Another related research may consider the real-time forecasting systems where the models automatically update based on the current market situation. The application of reinforcement learning for decision-making in stock management can be very insightful for enhancing sales prediction.

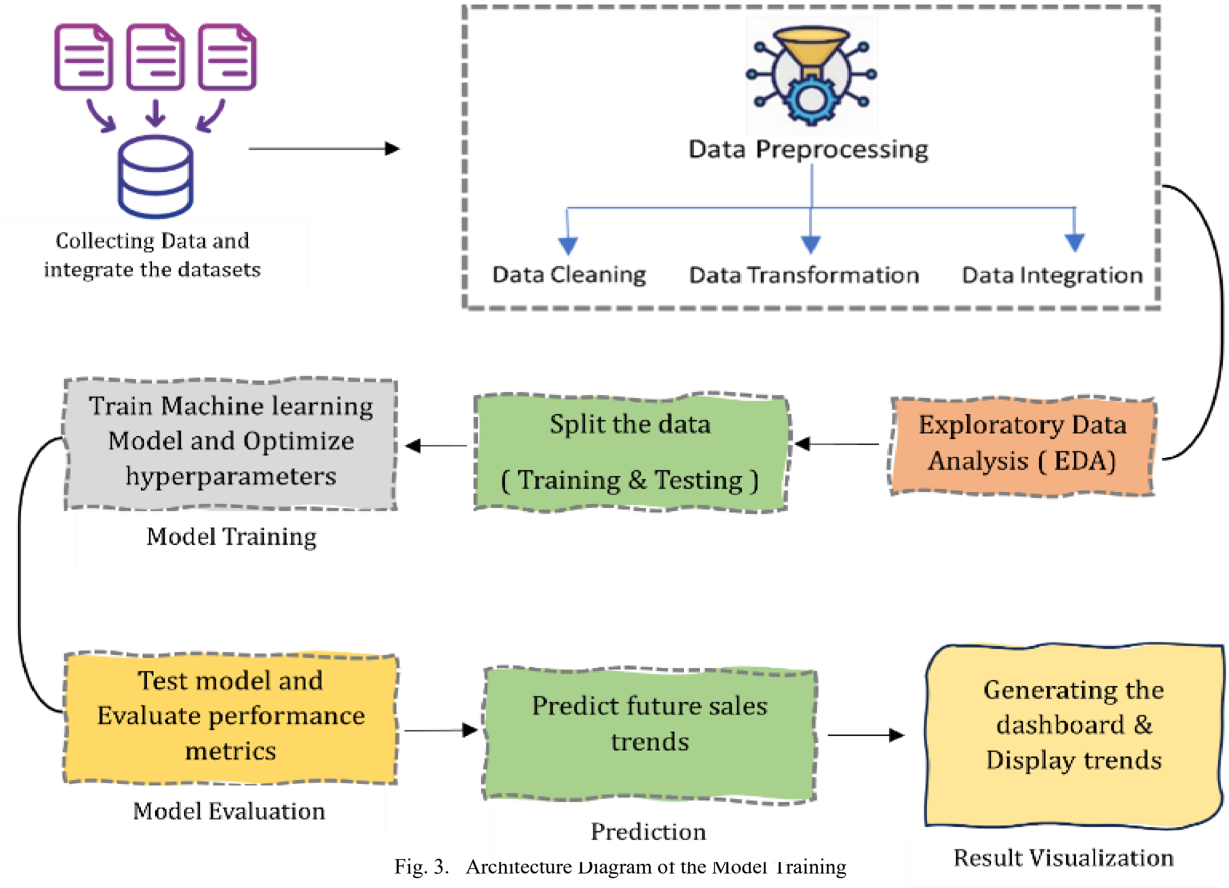

Fig. 3. Architecture Diagram of the Model Training

Conclusion

This paper evaluates the essential machine learning algorithms Decision Trees, Linear Regression, and Ridge Regression highlighting their programs, strengths, and barriers in real-global eventualities. Decision Trees include interpretability for classification and regression tasks in addition to simple Linear Regression but decisions trees can have issues like overfitting and multicollinearity. Ridge regression is complementary to Linear Regression through stemming from the same area of interest and controlling overfitting through regularization, which make Ridge Regression suited to high-dimensional data. Subsequent studies can pay more attention to the improvement for these models, including better regularization approaches and the construction of new blended versions for higher accuracy and feasibility in the realistic processes.

Authorized licensed use limited to: Mukesh Patel School of Technology & Engineering. Downloaded on July 10,2026 at 15:48:02 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] García, S., Ramírez-Gallego, S., Luengo, J., Benítez, J. M., & Herrera, F. (2016). Big data preprocessing: methods and prospects. *Big data analytics*, *1*, 1-22.

[2] Famili, A., Shen, W. M., Weber, R., & Simoudis, E. (1997). Data preprocessing and intelligent data analysis. *Intelligent data analysis*, *1*(1), 3-23.

[3] Jonckers, C. (2022). *The Application of Clustering Methods within the GLM Pricing Process* (Doctoral dissertation, Tilburg University).

[4] Behera, G., & Nain, N. (2020). A comparative study of big mart sales prediction. In *Computer Vision and Image Processing: 4th International Conference, CVIP 2019, Jaipur, India, September 27–29, 2019, Revised Selected Papers, Part I 4* (pp. 421-432). Springer Singapore.

[5] Thomassey, S., & Happiette, M. (2007). A neural clustering and classification system for sales forecasting of new apparel items. *Applied Soft Computing*, *7*(4), 1177-1187.

[6] van Ruitenbeek, R. E., Koole, G. M., & Bhulai, S. (2023). A hierarchical agglomerative clustering for product sales forecasting. *Decision Analytics Journal*, *8*, 100318.

[7] Wu, X., Kumar, V., Ross Quinlan, J., Ghosh, J., Yang, Q., Motoda, H., ... & Steinberg, D. (2008). Top 10 algorithms in data mining. *Knowledge and information systems*, *14*, 1-37.

[8] Müllner, D. (2011). Modern hierarchical, agglomerative clustering algorithms. *arXiv preprint arXiv:1109.2378*.

[9] Murtagh, F., & Contreras, P. (2012). Algorithms for hierarchical clustering: an overview. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, *2*(1), 86-97.

[10] Day, W. H., & Edelsbrunner, H. (1984). Efficient algorithms for agglomerative hierarchical clustering methods. *Journal of classification*, *1*(1), 7-24.

[11] Ahmed, M., Seraj, R., & Islam, S. M. S. (2020). The k-means algorithm: A comprehensive survey and performance evaluation. *Electronics*, *9*(8), 1295.

[12] Charbuty, B., & Abdulazeez, A. (2021). Classification based on decision tree algorithm for machine learning. *Journal of Applied Science and Technology Trends*, *2*(01), 20-28.

[13] Suthaharan, S. (2016). Machine learning models and algorithms for big data classification. *Integr. Ser. Inf. Syst*, *36*, 1-12.

[14] Maulud, D., & Abdulazeez, A. M. (2020). A review on linear regression comprehensive in machine learning. *Journal of Applied Science and Technology Trends*, *1*(2), 140-147.

[15] Liang, Y., Wu, J., Wang, W., Cao, Y., Zhong, B., Chen, Z., & Li, Z. (2019, August). Product marketing prediction based on XGboost and LightGBM algorithm. In *Proceedings of the 2nd international conference on artificial intelligence and pattern recognition* (pp. 150-153).

[16] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information processing systems*, *30*.

[17] Xia, Z., Xue, S., Wu, L., Sun, J., Chen, Y., & Zhang, R. (2020). ForeXGBoost: passenger car sales prediction based on XGBoost. *Distributed and Parallel Databases*, *38*, 713-738.

[18] Massaro, A., Panarese, A., Giannone, D., & Galiano, A. (2021). Augmented data and xgboost improvement for sales forecasting in the large-scale retail sector. *Applied Sciences*, *11*(17), 7793.

[19] Akande, Y. F., Idowu, J., Misra, A., Misra, S., Akande, O. N., & Ahuja, R. (2021, October). Application of XGBoost Algorithm for Sales Forecasting Using Walmart Dataset. In *International Conference on Advances in Electrical and Computer Technologies* (pp. 147-159). Singapore: Springer Nature Singapore.

[20] Chen, T., & Guestrin, C. (2016, August). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining* (pp. 785-794).

[21] Alice, K., & Jha, S. (2023). Sales Forecasting Based on Ensemble Learning. *Authorea Preprints*.

Authorized licensed use limited to: Mukesh Patel School of Technology & Engineering. Downloaded on July 10,2026 at 15:48:02 UTC from IEEE Xplore. Restrictions apply.